

LeaseOS Specialized Freight & Movement Compliance

Build Plan v1 - highway, specialty freight, over-dimensional movement, escort-provider integration, secure custody and evidence workflows

Design principle: LeaseOS should determine the applicable rule packs from the vehicle, cargo, route, jurisdiction and job - then fail closed when a legally or operationally required condition is unknown, expired, missing or contradicted. Third-party service providers remain interchangeable and are not hard-coded to any named company.

1. Core Architecture

Component	Build requirement
Specialty profile registry	Versioned rule packs activated by cargo, vehicle, movement type, jurisdiction and permit conditions.
Stackable classifications	A trip can simultaneously be TDG + tanker + overweight + overheight + secure-custody, rather than forcing one category.
Pre-departure gate	Evaluate driver, unit, trailer, cargo, documents, permits, route, weather/road restrictions, securement and required support services.
Authority hierarchy	Permit-specific conditions and current authoritative rules override generic LeaseOS guidance. Store source, effective dates and jurisdiction.
Evidence ledger	Timestamped actor, GPS, document/version, photos where appropriate, signatures, inspection events, corrections and reason codes.
Exception engine	Missing/expired/unknown/contradictory items become visible exceptions; safety-critical exceptions block dispatch.

2. Specialty Freight Rule Packs

Fuel / petroleum tanker

TDG classification and shipping document linkage; tank/trailer identity and inspection status; placard validation; loading/unloading checklist; grounding/bonding or site-required controls; compartment/load record; spill/emergency workflow; seal and custody events where used.

Propane, compressed gas and cryogenic

Pressure-vessel/trailer documentation; commodity-specific TDG controls; fill/load records; unloading connection checklist; emergency information; temperature/pressure telemetry interfaces where equipment supports them.

Refrigerated / temperature-controlled

Cargo-specific setpoint; pre-cool; continuous temperature history; door events; defrost events; sensor calibration status; sanitation records; temperature excursion workflow; receiver acceptance/rejection and cold-chain evidence report.

Dangerous goods general

UN number, shipping name, class/division, subsidiary class, packing group where applicable, quantity/container type, shipping document, placards/marks, training credential status, emergency-assistance requirements and jurisdictional validation.

Explosives / Class 1

Dedicated high-security compliance profile. Validate applicable authorization, classification/compatibility information, documentation, custody, authorized receiver and other legally required movement conditions. Keep sensitive operational information role-restricted.

Firearms / controlled or high-value freight

Commercial freight custody profile: authorized parties, package/pallet counts, seal/tamper events where used, receiver verification, proof of delivery and strict role-based visibility. Ammunition is separately evaluated for applicable TDG requirements.

Radioactive / regulated sources

Dedicated regulated-material rule pack with authority-specific documentation, package/category fields, trained personnel validation, custody and emergency-response information. No generic rule should substitute for the applicable regulator.

Lithium batteries / EV battery packs

Battery chemistry/type and condition, packaging status, damaged/defective exception, TDG classification where applicable, temperature/fire exception workflow and carrier documentation.

Waste / vacuum / oilfield fluids

Waste stream classification, generator, origin, quantity, manifest/ticket, approved destination, load/unload custody, partial acceptance, rejection, disposal receipt and complete generator-to-facility chain of custody.

Livestock

Species/load data, welfare/rest/water/ventilation requirements by jurisdiction, condition checks, delays, transfer-of-care records and exception escalation.

Logging / pipe / coils / precast / machinery

Commodity-specific securement templates, geometry/weight, blocking/bracing, tiedown/WLL evidence, edge protection, load-shift inspections and route constraints.

High-value / sensitive cargo

Need-to-know manifest visibility, custody scans, tamper exceptions, authorized receiver, proof of delivery and audit trail without exposing sensitive detail on ordinary dispatch/map views.

3. Heavy & Over-Dimensional Movement Engine

Movement dimension	LeaseOS planning fields / gates
Overwidth	Loaded width, permit thresholds/conditions, lane/route constraints, escort requirements, signs/lights/flags as required by authority
Overheight / high load	Loaded height, verified clearance data, unknown-clearance block, overhead utility/signal/structure conflicts, high-load support wo
Overlength / overhang	Overall combination length, load length, front/rear projection, turning constraints, permit/escort conditions.
Overweight	GVW plus individual axle and axle-group loads; permit limits; road/bridge/seasonal restrictions; route-specific evaluation.
Special configurations	Mobile cranes, service rigs, long combination vehicles, multi-axle heavy haul, steering dollies/boosters/jeeps and other configurat
Travel restrictions	Permit time windows, daylight/night restrictions, weather/visibility conditions, road bans, construction, municipal restrictions and j

4. Cargo Securement Engine

Securement should be a reusable calculation-and-evidence subsystem, not a generic checkbox. Capture cargo type, pieces, mass, dimensions, center-of-gravity information when known, anchor points, tiedown type/count, working load limits, blocking/bracing, dunnage, friction/anti-slip material and edge protection. Commodity-specific rules remain authoritative.

Workflow	Implementation
Plan	Generate a securement plan from cargo + vehicle + applicable commodity rule pack.
Verify	Driver confirms actual tiedowns/anchors and records exceptions; required evidence can be attached.
Reinspect	Rule engine schedules required en-route inspections based on the applicable jurisdiction/rules rather than a hard-coded universal dist
Exception	Shift, damaged tiedown, broken seal, compromised packaging or changed load geometry opens a corrective workflow.
Audit	Preserve original plan, inspection events, corrections, actor and reason instead of silently overwriting.

5. Pilot / Escort Vehicle Program

LeaseOS does not need to supply escort services. Build the operational parameters and an open third-party provider interface. No provider name is assumed or embedded.

- Determine escort requirement from permit conditions, movement dimensions, route and jurisdiction; store front/rear positioning and any additional authority-directed support.
- Provider qualification object: service region, credentials/qualifications where applicable, insurance, equipment capabilities, availability and verification dates.
- Assignment package: permit, route/version, dimensions, movement window, communication plan, meeting point and authorized operational contacts - only the information the provider needs.
- Trip state: requested -> accepted -> mobilizing -> staged -> active -> complete -> disputed/cancelled, with timestamps and evidence.
- Communication check before movement and event reporting for route blockage, clearance concern, breakdown, weather deterioration or permit-condition conflict.
- Provider cannot edit LeaseOS regulatory rules. Field feedback becomes a review request handled through the compliance-content change process.
- Service-provider billing remains separate from LeaseOS subscription/platform billing; support configurable referral/platform/service-management fee models later.

6. Third-Party Compliance / Field Validation Partner

Create an unnamed, non-exclusive **Service Partner** role. A future trucking-safety, pilot/escort or compliance company can audit workflows, perform field validation or deliver contracted services without becoming the authority that writes LeaseOS rules.

Government/permit sources remain the compliance authority; partner findings enter a controlled review and approval workflow.

7. Suggested Data Model

Table / object	Purpose
specialty_profiles	profile_id, code, name, version, jurisdictions, effective_from/to, status
trip_specialty_profiles	trip_id, profile_id, activation_reason, evaluated_at
cargo_items	trip/load, category, description, quantity, mass, dimensions, regulated identifiers, sensitivity level
securement_plans	load, rule_version, anchors, tiedowns, WLL data, blocking/bracing, verification status
securement_inspections	plan, timestamp, location, result, exceptions, actor, evidence
movement_dimensions	width, height, length, projections, GVW, axle/axle-group configuration
movement_permits	authority, permit number, effective period, conditions, route/version, document
clearance_checks	route segment/object, required clearance, known clearance, provenance, status
escort_requirements	movement, requirement source, count, positioning, police/utility/support flags
service_providers	tenant-independent provider record, regions, capabilities, qualification state
provider_assignments	job/trip, provider, scope, status, SLA, acceptance and completion evidence
custody_events	cargo/load, from/to party, timestamp, location, seal/tamper state, proof
telemetry_streams	temperature/pressure/door/etc. sensor identity, calibration and observations
compliance_exceptions	severity, gate impact, source, affected object, remediation, resolution
rule_sources	authority, URL/document identifier, jurisdiction, version/effective date, verification/review dates

8. Dispatch Gate Sequence

1. Resolve tenant, job, driver, unit, trailers and actual cargo.
2. Classify applicable specialty profiles; allow multiple simultaneous profiles.
3. Resolve jurisdiction(s), route, permit conditions and current versioned rule sources.
4. Validate driver/unit/trailer credentials and inspections.
5. Validate cargo documentation, TDG/regulated-material requirements and custody requirements where applicable.
6. Evaluate dimensions, axle groups, road/bridge/clearance constraints and permit conditions.
7. Validate securement plan and required inspection schedule.
8. Determine required escort/pilot/utility/police or other support from authoritative conditions.
9. Evaluate road/weather/time-window restrictions and offline-data freshness.
10. Return READY, READY_WITH_ACKNOWLEDGED_NONSAFETY_EXCEPTIONS, or BLOCKED. Safety-critical unknowns are never treated as safe.
11. At departure, freeze a dependency fingerprint of the rule versions, permit, route and evidence used for the decision.

9. Implementation Checkpoints

Checkpoint	Deliverable	Gate
S1	Specialty profile registry + stackable classification	Pure tests for deterministic profile activation.
S2	Heavy/over-dimensional schema + permit-condition parser	Unknown bridge/clearance must fail closed.
S3	Securement plan + inspection ledger	No silent overwrite; correction/audit tests.
S4	Fuel/TDG + reefer rule packs	Versioned authority source attached to every compliance decision.
S5	Waste/custody + sensitive/high-value profiles	RBAC and tenant isolation tests.
S6	Escort requirement + open provider interface	No named/provider-specific coupling; provider cannot alter rules.

Checkpoint	Deliverable	Gate
S7	Telemetry adapters + offline queue	Loss of network does not lose observations/events.
S8	Dispatch-gate integration	Regression suite proves ordinary freight is not unnecessarily blocked.
S9	Field validation sandbox	Partner feedback creates review requests, never production rule edits.
S10	Production readiness	Legal/regulatory review, security review, real-device/offline tests, source freshness monitoring.

10. Important Product Boundaries

- LeaseOS should not claim that a calculated route, securement plan or permit interpretation is itself legal authorization. The applicable authority and issued permit remain controlling.
- Do not hard-code example temperatures, escort thresholds, securement distances, dimensions or weights as universal legal rules. They vary by cargo, equipment, jurisdiction, permit and rule version.
- Sensitive cargo information should use least-privilege access, encrypted storage/transport, immutable audit events and minimized map/dispatch display.
- Regulatory data needs source provenance, effective dates, review dates and a stale-data state. Unknown or stale safety-critical data cannot silently become 'clear'.
- Third-party validation is valuable but does not replace required regulatory, legal, engineering or authority approval.

11. Relationship to Existing LeaseOS Plan

This package extends the existing LeaseOS mapping/routing, provenance, confirmation, audit, offline-first and dispatch-gate architecture. It should be implemented after the core stabilization and tenant-scoping work, then integrated into the same canonical gateway rather than built as a parallel application.